



SOUTHEAST UNIVERSITY

BOOK NEST

LIBRARY MANAGEMENT SYSTEM

PROJECT REPORT

Course Title : Programming Language II Lab (Java)

Course Code : CSE282

Section: 6

Department : CSE

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1. Introduction

Overview

Book Nest is a Java-based Library Management System developed to organize common library activities such as managing books, maintaining stock, borrowing and returning books, and recording purchases. The system also provides user registration and login functionality. The project applies Object-Oriented Programming concepts and uses Java file serialization to store users, books, and purchase records so that the data can be loaded again when the application is reopened.

2. Objectives

What the Project Aims to Achieve

- Develop a simple and organized library management system.
- Manage book information and stock efficiently.
- Provide user registration and login functionality.
- Support book borrowing, returning, and purchasing.
- Store user, book, and purchase data using file handling.
- Demonstrate the four fundamental OOP principles in Java.

3. Motivation

Why We Selected This Project

Libraries need an organized way to maintain book records, stock information, and transaction records. Manual management can make searching and updating records time-consuming. We selected this project because it provides a practical real-world problem through which Java programming and OOP concepts can be applied to a complete application.

4. Applications

Where the Project Can Be Used

- University and college libraries
- School libraries
- Small institutional libraries
- Personal or community book collections

Potential Users

- Library readers/users
- Library administrators
- Library staff

Practical Uses and Benefits

- Faster book searching and updating
- Easier stock management
- Simple borrowing and returning process
- Organized purchase records
- Persistent storage of important records

5. Project Description

Major Modules / Classes

Java Class	Main Responsibility
Main	Controls program flow and displays the main menus.
User	Abstract class containing common user information.
Reader	Represents a regular library user.
Admin	Represents an administrative user and contains an admin menu.
Book	Stores book details and quantity.
Purchase	Stores purchase information.
Library	Performs book management and transaction operations.
FileManager	Saves and loads users, books, and purchases.
LibraryGUI	Provides Swing-based login and registration.

Workflow of the System

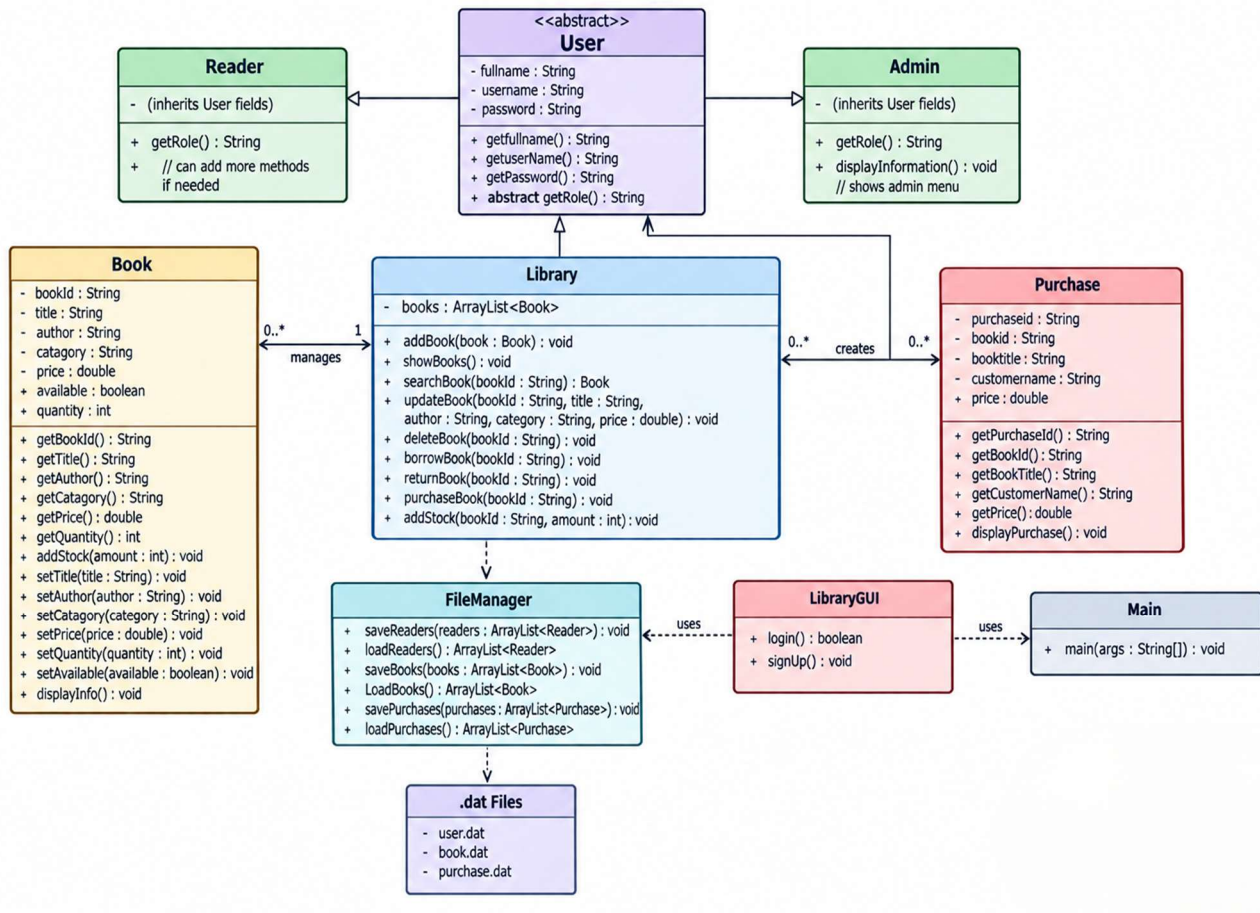
1. The application starts from Main.
2. The user can choose Login, Sign Up, or Exit.
3. Sign Up creates a Reader account and saves it to user.dat.
4. Login checks the entered credentials against saved Reader accounts.
5. After successful login, the Library Management System menu is displayed.
6. The user selects a book operation such as add, search, update, delete, borrow, return, purchase, or add stock.
7. Library performs the selected operation and FileManager saves updated data when required.

Java / OOP Concepts Used

- Classes and Objects
- Constructors
- Encapsulation
- Abstraction
- Inheritance
- Polymorphism / Method Overriding
- ArrayList

- Methods and Parameters
- Exception Handling
- File Handling and Serialization
- Java Swing

Class Diagram



6. Tools & Technologies

Tool / Technology	Use
Programming Language	Java
IDE	Visual Studio Code
GUI	Java Swing / JOptionPane
Collection	ArrayList
Persistence	Java File I/O and Object Serialization
Data Files	user.dat, book.dat, purchase.dat
Version Control	Git / GitHub

7. Output / Results

Screenshots of the Developed System

Main Menu

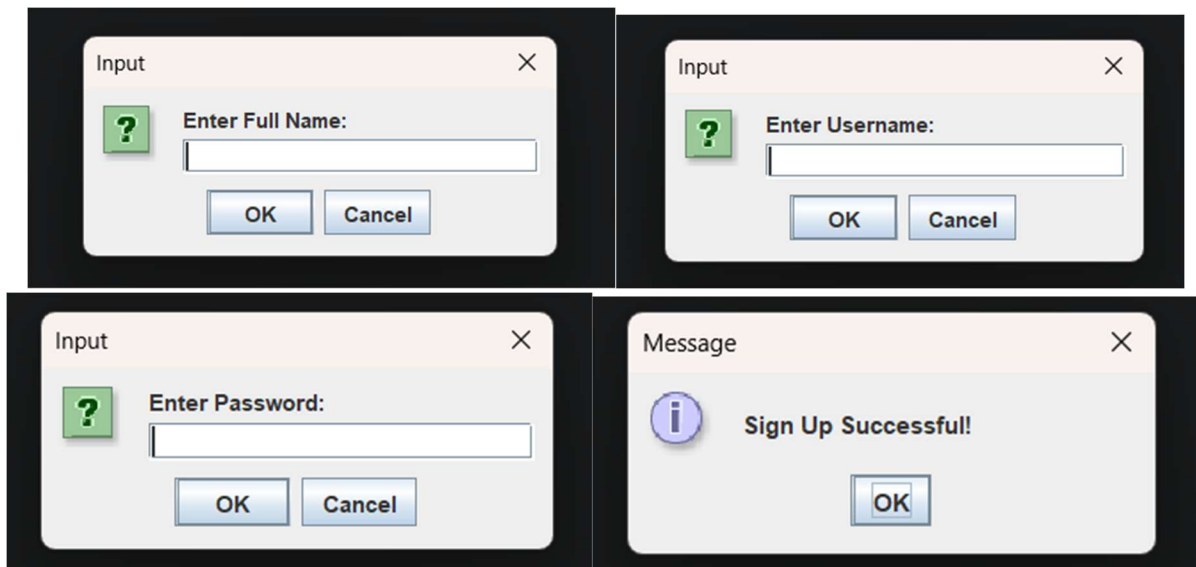
```
MINGW64 ~/Book-Nest (aurpy)
$ java Main

===== Book Nest =====

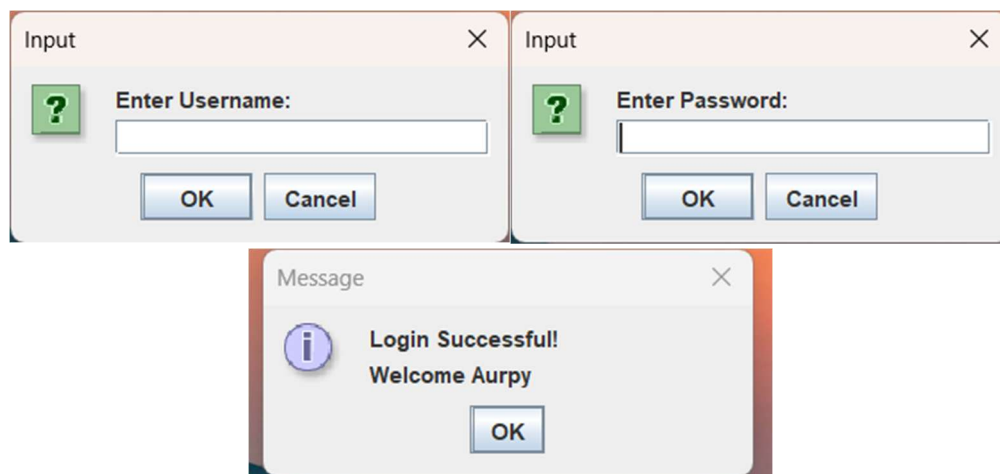
1. Login
2. Sign Up
3. Exit

Enter your choice :
|
```

Sign up



Log in



LIBRARY MANAGEMENT MENU

```
===== Library Managment System =====

1. Add Book
2. Show Books
3. Search Book
4. Update Book
5. Delete Book
6. Borrow Book
7. Return Book
8. Purchase Book
9. Add Stock
10. Exit

Enter your choice: █
```

ADD BOOK

```
Enter your choice: 1
Enter Book ID :
104
Enter Title :
The Secret of Secrets
Enter Author :
Dan Brown
Enter Category :
Novel
Enter Price :
600
Enter Quantity :
1
Books saved successfully.
Book added Successfully!
```

SHOW / SEARCH BOOK

```
Enter your choice: 2
Book ID: 101
Title: To Kill a Mockingbird
Author: Harper Lee
Category: Fiction
Price: 699.0
Quantity: 19
-----
Book ID: 102
Title: A Brief History of Time
Author: Stephen Hawking
Category: Science
Price: 550.0
Quantity: 19
-----
Book ID: 103
Title: Atomic Habits
Author: James Clear
Category: Self-Help
Price: 499.0
Quantity: 10
-----
```

```
Enter your choice: 3
Enter Book Id :
102
Book ID: 102
Title: A Brief History of Time
Author: Stephen Hawking
Category: Science
Price: 550.0
Quantity: 19
```

UPDATE / DELETE BOOK

```
Enter your choice: 4
Enter Book ID :
104
Enter Title :
The Secret of Secrets
Enter Author :
Dan Brown
Enter Category :
Novel
Enter Price :
700
Books saved successfully.
Book updated successfully.
```

```
Enter your choice: 5
Enter Book ID: 105
Books saved successfully.
Book deleted successfully.
```

BORROW / RETURN BOOK

```
Enter your choice: 6
Enter Book ID: 104
Books saved successfully.
Book borrowed successfully.
Remaining stock: 0
```

```
Enter your choice: 7
Enter Book ID: 104
Books saved successfully.
Book returned successfully.
Current stock: 1
```

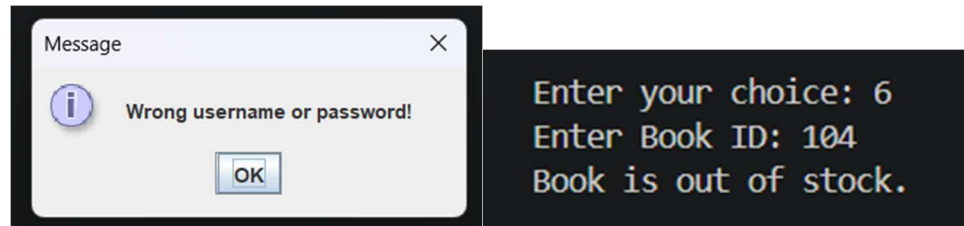
PURCHASE BOOK

```
Enter your choice: 8
Enter Book ID: 104
Enter Customer Name: Aurpy
Books saved successfully.
Stock reduced.
Remaining stock: 0
Purchases saved successfully.

Purchase successful!

===== Purchase Information =====
Purchase ID   : P113
Book ID      : 104
Book Title   : The Secret of Secrets
Customer Name : Aurpy
Price       : 700.0
```

Error Handling / Output



8. Contribution

All four group members worked together throughout the development of the project. We collaboratively contributed to the coding, testing, debugging, documentation, and overall implementation of the system.

9. Conclusion

Summary of the Completed Project

Book Nest was developed as a Java-based Library Management System to organize book records and common library operations. The completed implementation includes user registration and login, book CRUD operations, stock management, borrowing, returning, purchasing, and file-based data persistence.

Limitations of the Project

- The current main library interface is primarily console-based, while Swing is used for login and registration.

- The project does not currently use a database such as MySQL or SQLite.
- Custom project-specific exception classes are not included in the current implementation.
- The current login flow authenticates Reader accounts; full role-based Admin authentication can be expanded.

Possible Future Improvements

- Develop a complete Swing-based library dashboard.
- Add separate Reader and Admin interfaces with role-based access.
- Add category, title, and author-based searching.
- Add purchase history for individual users.
- Introduce custom exception classes.
- Improve password security and authentication.
- Add book cover images and a more advanced graphical interface.